

Non-Interfering Posttraining: Geometric Skill Isolation and Contrastive Self-Distillation for Capability-Preserving Alignment

Project Description

Project Summary

Overview. Reinforcement-learning posttraining (PPO, GRPO, and their variants) has become the standard route to stronger reasoning in language models [1–3], but the same update that sharpens a target capability also acts as a high-gain behavioral intervention. Recent work shows that even a single biased GRPO example can induce systematic stereotype-driven reasoning across fairness benchmarks [4], and related analyses of posttraining and self-distillation show that dense optimization signals can also shift reasoning style, uncertainty expression, and prior behavior in unintended ways [5]. Current remedies treat the symptom after the fact, either by adding a KL penalty toward a frozen reference or by running a second supervised repair stage. This project asks a more basic question: can posttraining be designed so that improving one skill does not, by construction, disturb the others? We propose two complementary mechanisms. The first, *Skill-Orthogonal Optimization*, extends Muon-style spectral update orthogonalization [6–9] so that the update for a target skill is constrained to the orthogonal complement of a protected skill subspace, preventing collateral movement before it happens. The second, *Contrastive On-Policy Self-Distillation (C-OPSD)*, builds on on-policy distillation and self-distillation [10–12] while incorporating the distributional insight from *Shape of Thought* that model-proximal reasoning traces can be useful even when final answers are not uniformly correct [13]. It also connects to related work by Chandra and collaborators on using model hierarchies to improve evaluation and reasoning [14]. C-OPSD replaces scalar reward with a denser distributional signal by distilling from both correct and incorrect self-generated trajectories, pulling the policy toward verified-correct token distributions and pushing it away from verified-incorrect ones. Together they form a non-interfering posttraining framework that is cheap, teacher-free, and reproducible on single-node academic hardware. The work introduces a geometric account of why posttraining causes behavioral drift and a tractable optimizer-level intervention (subspace-constrained orthogonalization) that addresses it at the source rather than through post-hoc regularization. It also generalizes on-policy self-distillation from a positive-only objective to a contrastive one that makes principled use of negative (incorrect) examples, informed by prior findings on on-policy distillation, self-distillation stability, and the value of distribution-matched reasoning traces [5, 10, 11, 13]. The two mechanisms are studied independently and jointly, giving a clean ablation of geometry versus supervision signal.

Broader Impacts. Bias amplification during alignment is a documented safety risk for models deployed in hiring, education, and public-facing question answering. A posttraining method that preserves fairness and knowledge priors while improving reasoning directly reduces that risk. Because the approach is teacher-free and adapter-compatible, it lowers the compute barrier for academic and under-resourced groups doing alignment research, and it repurposes existing process-annotation datasets as reusable supervision. The project supports graduate and undergraduate training, open-source releases of code and small-model checkpoints, and public documentation of both successes and failure modes.

1. Introduction and Motivation

Posttraining is where a general pretrained model becomes a useful one. It is also where a surprising amount of unintended behavior is introduced. The dominant tools, GRPO and its relatives, normalize rewards within a sampled group and update the policy by relative advantage, which makes them memory-efficient and effective for verifiable reasoning tasks [2, 3]. The same update is a concentrated behavioral edit. If the reward credits

a spurious reasoning pattern, GRPO reinforces the pattern, not just the correct answer. A recent one-shot result demonstrated that a single biased training item can produce systematic stereotype-driven generalization across attributes and fairness benchmarks [4], which moves GRPO from a purely capability-improving tool to a safety-relevant optimization surface.

Current best practices manage this problem downstream. In-loop KL anchoring periodically distills the trainable policy toward a frozen reference under the same context, and sequential repair (GRPO followed by supervised distillation) applies a corrective pass after the reinforcement stage. Both help, and both share a limitation. They treat interference as something to counteract with an added penalty or an extra stage, after the update geometry has already coupled the skills together. The reference is asked to pull the model back, which means the pull is always in tension with the reward that pushed it away.

We take a different starting point. Distinct skills and behaviors induce partially separable directions in a model’s local update geometry, visible through gradients, activations, or low-rank adapter subspaces. Interference happens when an update that improves a target skill also has large components along directions that are predictive of protected behaviors. Instead of asking a frozen reference policy to pull the model back after each step, we ask whether the step itself can be made locally non-interfering: first remove protected components in a chosen update representation, and only then apply the optimizer’s conditioning or orthogonalization inside the remaining subspace. This keeps the proposal close to the geometric intuition behind orthogonal gradient descent in continual learning [15, 16], while treating Muon-style Newton–Schulz orthogonalization [6, 8, 9] as one possible conditioning operator rather than as the source of the non-interference guarantee.

A second observation concerns supervision density. GRPO reduces every sampled completion to a scalar reward, discarding most of the information in the trajectory. Self-distillation methods instead match full token distributions over on-policy rollouts [10–12], but the standard formulation learns only from correct behavior. In reasoning, the informative signal often lives at the branch points where a correct trace and an incorrect trace diverge; this is consistent with evidence that the distributional shape of reasoning traces can matter independently of final-answer correctness [13]. A method that annotates both and contrasts them can supervise exactly those decision tokens, which is denser than reward and more targeted than uniform distillation.

These two ideas, geometry and contrastive supervision, attack the same failure from opposite ends. This proposal develops both and studies whether, together, they yield posttraining that improves a target capability without eroding the behavioral prior already present in the model.

2. Research Aims

2.1 Aim 1: Subspace-Conditioned Posttraining Updates

The first aim is not to claim that Muon alone can preserve unrelated behaviors. Muon orthogonalizes a momentum matrix $M \in \mathbb{R}^{m \times n}$ using a Newton–Schulz approximation to its polar factor, producing a spectrally balanced update direction [6, 8, 9]. This conditioning can change the shape of an update, and therefore a naive “project the gradient, then run Muon” formulation is not obviously safe for arbitrary protected directions. We therefore frame the method more abstractly as *subspace-conditioned posttraining*: a protected component is removed in a local update representation, and the optimizer is allowed to condition only the remaining component.

First, we define a *protected update subspace* for each behavior we wish to preserve. Given small anchor batches for a protected skill (for example a fairness set, a general-knowledge set, or an instruction-following set), we collect reference gradients, activation-gradient products, or adapter updates, and extract a low-rank orthonormal basis for the directions that most affect that behavior. In a flattened representation this gives a

basis $P \in \mathbb{R}^{d \times r}$ and projector

$$Q = I - PP^\top. \quad (1)$$

For a layerwise matrix update, the same idea can be represented through left and right projectors,

$$Q_L = I - P_L P_L^\top, \quad Q_R = I - P_R P_R^\top, \quad (2)$$

where P_L and P_R capture protected output- and input-side directions. This follows the subspace-protection logic of orthogonal gradient descent [15, 16], but the protected directions are estimated from model behaviors relevant to posttraining rather than from a sequence of supervised tasks.

Second, we apply a *project-then-condition* update. In the flattened case the target update g is first projected as

$$\tilde{g} = Qg, \quad (3)$$

and then passed to an optimizer-specific conditioning operator $\mathcal{C}$, with an optional final projection for safety:

$$\Delta\theta = Q\mathcal{C}(\tilde{g}). \quad (4)$$

For ordinary Adam-like updates, $\mathcal{C}$ is diagonal preconditioning. For Shampoo-like updates, $\mathcal{C}$ is a Kronecker/tensor-factored second-order preconditioner [17, 18], which is a more direct precedent for structure-aware preconditioning than Muon. For Muon-like updates, $\mathcal{C}$ is Newton–Schulz/polar conditioning applied after subprojection. In the layerwise matrix case, we use

$$\tilde{M} = Q_L M Q_R, \quad \Delta W = Q_L \mathcal{C}(\tilde{M}) Q_R. \quad (5)$$

The non-interference property comes from the explicit projectors; Muon is only used to improve the conditioning of the allowed update. If the protected space is represented as a row, column, or adapter subspace, the projection is structurally preserved by applying the final Q_L, Q_R projection. For arbitrary flattened protected directions, the final projection is essential because nonlinear conditioning can otherwise reintroduce forbidden components.

This gives a conservative first-order guarantee: the applied update is orthogonal to the protected subspace in the chosen local representation,

$$P^\top \Delta\theta = 0 \quad \text{or} \quad P_L^\top \Delta W = 0, \quad \Delta W P_R = 0, \quad (6)$$

so protected behavior should move less under a local linear approximation. The proposal does not assume this completely freezes behavior, since model behavior is nonlinear and the protected basis is only approximate. Instead, the research question is empirical and geometric: which representation of the protected subspace gives the best target-gain versus drift tradeoff, and when does Muon-style conditioning help after the safe subprojection?

A parameter-efficient variant applies the same principle to per-skill LoRA adapters [19]. Rather than claiming that one global optimizer step can isolate all behaviors, we learn separate low-rank update spaces and discourage overlap between them,

$$\mathcal{L}_{\text{orth}} = \sum_{i \neq j} \|A_i^\top A_j\|_F^2, \quad (7)$$

with optional projection of each new adapter update away from previously learned protected adapter subspaces. This makes the aim an ablation over increasingly structured variants: projection-only, projection plus Adam-style diagonal preconditioning, projection plus Shampoo-style Kronecker/factored preconditioning, and projection plus Muon-style spectral conditioning. The hypothesis is that most protection comes from the subspace constraint, while second-order or spectral conditioning may improve optimization inside the allowed subspace.

2.2 Aim 2: Contrastive On-Policy Self-Distillation (C-OPSD)

Standard on-policy self-distillation samples completions from the current policy, verifies them, and distills the policy toward the correct ones [11]; more broadly, on-policy distillation addresses the distribution mismatch created when students are trained only on fixed off-policy sequences [10]. C-OPSD keeps the on-policy sampling but uses the incorrect completions as an explicit negative signal.

At each step we sample a group of completions per prompt and label each correct or incorrect, cheaply through answer verification for reasoning tasks and through existing annotations otherwise. Correct trajectories provide a positive distillation target, and we distill sparsely, supervising the tokens where correct and incorrect traces diverge rather than every token, which concentrates the signal on the reasoning decisions that matter; this design is motivated both by process-supervision data such as PRM800K [20] and by findings that distribution-matched reasoning traces may remain useful even when imperfect [13]. Incorrect trajectories provide a repulsion term implemented as a bounded contrastive log-ratio between matched correct completions y^+ and incorrect completions y^- on the same prompt x ,

$$\mathcal{L}_{\text{C-OPSD}} = \underbrace{\mathbb{E}_{y^+} \text{D}_{\text{KL}}(\pi_0(\cdot | x) \| \pi_\theta(\cdot | x))}_{\text{positive distillation}} - \lambda \underbrace{\mathbb{E}_{(y^+, y^-)} \left[\max(0, m - (\ell_\theta(y^+) - \ell_\theta(y^-))) \right]}_{\text{bounded contrastive repulsion}}, \quad (8)$$

where ℓ_θ is the (length-normalized) sequence log-likelihood and m is a margin.

The choice of divergence is a first-class research question because self-distillation is sensitive to it. On-policy distillation work shows that training on the student’s own rollout distribution and the direction/form of the divergence matter [10–12], while recent analysis of self-distillation failures shows that distillation can suppress useful uncertainty and fork-exploration behavior [5]. The negative term is exactly where such collapse is most likely, so we bound the repulsion with a margin or clip and study which divergences remain stable when negatives are included. The hypothesis is that contrasting verified-wrong against verified-right supervises reasoning far more densely than scalar reward, without the external teacher that ordinary distillation assumes.

2.3 Aim 3: Integration and Analysis

We evaluate each mechanism alone and combined, since they are complementary. Aim 1 protects skills by constraining the local update subspace, and Aim 2 enriches the supervision signal. The integrated system runs C-OPSD as the learning objective while subspace-conditioned optimization governs how each update is applied. We ablate projection-only updates, projection plus Adam-style conditioning, and projection plus Muon-style spectral conditioning, so that any gain from Muon is measured rather than assumed. We also measure whether contrastive supervision and geometric protection are additive or whether one mechanism subsumes the other.

3. Evaluation Plan and Datasets

We adopt an evaluation suite that separates the target capability from the behaviors we wish to protect. For reasoning we use **GSM8K** [21] and **MATH** [22], both verifiable, so correct-versus-incorrect labels for C-OPSD come for free. For the contrastive objective specifically, **PRM800K** is a natural fit because it already contains step-level correct and incorrect annotations [20], which lets us reuse process-supervision data as a distillation signal rather than only as reward-model training data.

For the protected axes we use **MMLU** [23] for broad knowledge and a focused behavioral suite for stereotype and disparity drift: **BBQ** [24], **CrowS-Pairs** [25], **StereoSet** [26] (inter- and intra-sentence deviation from parity), **WinoBias** [27] (the pro-minus-anti gap), and **DiscrimEval** age disparity [28]. Lower is better on the

stereotype and disparity metrics. To test skill isolation beyond fairness, we add a code axis (**HumanEval** [29], **MBPP** [30]) and an instruction-following axis (**IFEval** [31]), so we can measure whether optimizing math leaves code and instruction adherence untouched.

Experiments run across Qwen-family instruction models at 0.5B, 3B, and 7B scales, matching recent GRPO stability studies [4] so our numbers are directly comparable to standard GRPO, in-loop KL anchoring, and sequential supervised repair. The primary outcome is a Pareto comparison: for a fixed improvement on the target axis, how much movement occurs on the protected axes. A successful method improves reasoning while holding fairness and knowledge near the reference.

4. Broader Impacts

Safer and fairer deployment. The failure mode this project targets, bias amplification during posttraining, matters most in the settings where models are increasingly used, including hiring support, educational tutoring, and public question answering [4, 28]. A method that preserves fairness priors while improving reasoning reduces the chance that a routine capability update silently degrades behavior in a deployed system.

Efficiency and access. Both mechanisms are teacher-free and compatible with parameter-efficient fine-tuning, so a full study fits on single-node hardware. This lowers the barrier for academic and under-resourced groups to do alignment research that currently assumes large teacher models or multi-stage pipelines. Repurposing existing process-annotation datasets (such as PRM800K) as distillation signal further stretches limited data budgets.

Reproducibility and open science. We will release code, training recipes, and small-model checkpoints, and we will document failure modes explicitly, including the divergence choices that cause collapse under contrastive self-distillation. Publishing what does not work is part of the contribution, since the stability of these objectives is poorly characterized.

Education and workforce. The project provides well-scoped research problems for graduate and undergraduate students spanning optimization theory, systems, and evaluation, and the small-scale reproducibility makes it suitable for classroom and hands-on training.

5. Timeline and Expected Outcomes

Year one develops Skill-Orthogonal Optimization, establishes the protected-subspace machinery, and reproduces the baseline suite at all three scales. Year two develops and stabilizes C-OPSD, characterizes the divergence choices, and integrates the two mechanisms. A final phase expands the fairness and skill axes and prepares open releases.

Expected outcomes are a geometric account and an optimizer-level method for capability-preserving post-training, a contrastive self-distillation objective that makes principled use of negative examples, a public benchmark comparison against standard GRPO and its stabilized variants, and open artifacts (code, recipes, checkpoints) reproducible on modest hardware. The unifying result we aim to establish is that posttraining can be designed to minimize interference at the update-construction stage, rather than relying only on repairs applied after drift has already occurred.

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